

$\mathbf{a}'_{\text{true}}$ arm, 4-state: the update half-step

(Companion of 0902; $\mathbf{x} = [\delta\bar{w}_1, \delta\bar{w}_2, \delta\bar{w}_3, \bar{a}_w]^\top$, $e = x - \hat{x}$; stages inside call k : $\hat{x}^-$ after predict, $\hat{x}^{(1)}$ after the y_1 update, $\hat{x}$ after the y_2 update; ℓ_{ij} = entry (i, j) of $L[k]$; hold: $\Delta\bar{w}_d = 0$, $\nabla_d\bar{w}_d = 0$, $F_{dw} = 0$)

1 Update half-step

$$\begin{aligned} S_1[k] &= P_{11}^-[k] + R_1, & \ell_{i1}[k] &= \frac{P_{i1}^-[k]}{S_1[k]} \\ e_{y_1}[k] &= y_1[k] - \hat{x}_1^-[k] = e_1^-[k] + n_w[k] \\ \hat{x}^{(1)}[k] &= \hat{x}^-[k] + \ell_{.1}[k] e_{y_1}[k] \\ e^{(1)}[k] &= e^-[k] - \ell_{.1}[k] (e_1^-[k] + n_w[k]) \\ P^{(1)}[k] &= (I - \ell_{.1}e_1^\top)P^-(I - \ell_{.1}e_1^\top)^\top + \ell_{.1}R_1\ell_{.1}^\top \end{aligned}$$

$$\begin{aligned} H_2[k] &= a_{\text{cov}}(1 - S) [0, 0, 0, 1] \quad (\text{hold}) \\ S_2[k] &= H_2P^{(1)}H_2^\top + R_2, & \ell_{.2}[k] &= \frac{P^{(1)}H_2^\top}{S_2} \\ e_{y_2}[k] &= y_2[k] - H_2\hat{x}^{(1)}[k] \\ \hat{x}[k] &= \hat{x}^{(1)}[k] + \ell_{.2}[k] e_{y_2}[k] \\ e[k] &= e^{(1)}[k] - \ell_{.2}[k] e_{y_2}[k] \end{aligned}$$

gain row

$$\hat{a}_w[k] - \hat{a}_w^-[k] = \ell_{41}[k] e_{y_1}[k] + \ell_{42}[k] e_{y_2}[k]$$

2 Lemma: $\ell_{41} = -\bar{a}' \ell_{31}$ in the hold

error dynamics of 0902 (hold, first order), and the law-consistent combination

$$\begin{aligned} e_3^-[k+1] &= \lambda_c e_3[k] - F_{dw}[k] e_4[k] - \varepsilon_w[k] \\ e_4^-[k+1] &= e_4[k] + \bar{a}'[k] [(1 - \lambda_c) e_3[k] + F_{dw}[k] e_4[k] + \varepsilon_w[k]] \\ e_4^-[k+1] + \bar{a}'[k] e_3^-[k+1] &= e_4[k] + \bar{a}'[k] e_3[k] \end{aligned}$$

$$\begin{aligned} \text{Cov}(e_4 + \bar{a}'e_3, e_3)^-[k+1] &= \lambda_c \text{Cov}(e_4 + \bar{a}'e_3, e_3)[k] - F_{dw}[k] \text{Cov}(e_4 + \bar{a}'e_3, e_4)[k] \\ \text{Cov}(e_4 + \bar{a}'e_3, e_1)^-[k+1] &= \text{Cov}(e_4 + \bar{a}'e_3, e_2)[k] \\ \text{Cov}(e_4 + \bar{a}'e_3, e_2)^-[k+1] &= \text{Cov}(e_4 + \bar{a}'e_3, e_3)[k] \end{aligned}$$

$$F_{dw} = 0 \Rightarrow \text{Cov}(e_4 + \bar{a}'e_3, e_1)^- \rightarrow 0 \Rightarrow P_{41}^- = -\bar{a}' P_{31}^- \Rightarrow \ell_{41} = -\bar{a}' \ell_{31}$$

(the combination $e_4 + \bar{a}'e_3$ is untouched by the predict and, once its covariance with e_1 has decayed as λ_c^k , by the y_1 update; $F_{dw} \neq 0$ sources it with $-F_{dw} \text{Var}(e_4 + \bar{a}'e_3)$, which is where $\ell_{41} + \bar{a}'\ell_{31} \neq 0$ in motion comes from; the y_2 update sources $\text{Cov}(e_4 + \bar{a}'e_3, e_j)$ by $-(H_{24}^2 P_{j4}^{(1)}/S_2) \text{Cov}(e_4 + \bar{a}'e_3, e_4)$, a factor $\sim 10^{-5}$, so the relation holds to that order; the decay of $\text{Cov}(e_4 + \bar{a}'e_3, e_1)$ is set by the predict-plus- y_1 closed loop, eigenvalue ≈ 0.41 , faster than λ_c ; log 2026-09-03: $|\ell_{41} + \bar{a}'\ell_{31}| < 10^{-4}$ in every hold, both trajectories)

3 State dependence of the gains

$$\begin{aligned}\bar{a}'[k] &= \bar{a}'_{\text{true}}(\bar{w}_d[k-1] - \delta\hat{w}_3[k-1]) = \bar{a}'_0 - \bar{a}''(\delta\hat{w}_3[k-1] - \mathbb{E}[\delta\hat{w}_3]) + \mathcal{O}(e^2) \\ \ell_{41}[k] &= -\bar{a}'[k]\ell_{31}[k] = \ell_{41,0} + \bar{a}''\ell_{31}(\delta\hat{w}_3[k-1] - \mathbb{E}[\delta\hat{w}_3]) \\ \ell_{42}[k] &= \frac{P_{44}^{(1)} H_{24}}{H_{24}^2 P_{44}^{(1)} + R_2(\hat{a}_w)}, \quad H_{24} = a_{\text{cov}}(1 - S(\hat{a}_w)), \quad R_2 = 2a_{\text{cov}}^2 IF_{\text{var}}(\hat{a}_w)(\hat{a}_w + \xi)^2 \\ \ell_{42}[k] &= \ell_{42,0} + \frac{\partial \ell_{42}}{\partial \hat{a}_w}(\hat{a}_w^{(1)}[k] - \mathbb{E}[\hat{a}_w]) \\ \frac{\partial \ell_{42}}{\partial \hat{a}_w} &= \frac{\ell_{42}}{H_{24}} \left(1 - \frac{2H_{24}^2 P_{44}^{(1)}}{S_2}\right) \frac{\partial H_{24}}{\partial \hat{a}_w} - \frac{\ell_{42}}{S_2} \frac{\partial R_2}{\partial \hat{a}_w}\end{aligned}$$

4 Process/measurement noise cross-covariance M

$$\begin{aligned}w[k] &:= x[k+1] - F[k]x[k] - B[k]u[k] = \varepsilon_w[k][0, 0, -1, \bar{a}'[k]]^\top \\ \varepsilon_w[k] &= \bar{a}_w[k]\bar{f}_T[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{a}_w[k-i]\bar{f}_T[k-i] + (1 - \lambda_c)n_w[k] \\ y_1[k] &= \delta\bar{w}_1[k] + n_w[k] \\ Q[k] &= \mathbb{E}[ww^\top] = Q_{33}[k][0, 0, -1, \bar{a}']^\top[0, 0, -1, \bar{a}'] \\ Q_{33} &= \kappa_T \left(\hat{a}_w + (1 - \lambda_c)^2 \sum_{i=1}^d \hat{a}_w[k-i] \right) + (1 - \lambda_c)^2 R_1 \\ M[k] &:= \mathbb{E}[w[k]n_w[k]] = R_1 g_n[k], \quad g_n[k] := (1 - \lambda_c)[0, 0, -1, \bar{a}'[k]]^\top \\ \frac{MM^\top}{R_1} &= (1 - \lambda_c)^2 R_1 [0, 0, -1, \bar{a}']^\top [0, 0, -1, \bar{a}'] \quad (\text{the } n_w \text{ share of } Q)\end{aligned}$$

5 Mean of the update half-step

$$\begin{aligned}\mathbb{E}[\hat{a}_w[k] - \hat{a}_w^-[k]] &= \ell_{41,0} \mathbb{E}[e_{y_1}[k]] + \ell_{42,0} \mathbb{E}[e_{y_2}[k]] \\ &\quad + \bar{a}''\ell_{31} \mathbb{E}[(\delta\hat{w}_3[k-1] - \mathbb{E}[\delta\hat{w}_3])e_{y_1}[k]] \\ &\quad + \frac{\partial \ell_{42}}{\partial \hat{a}_w} \mathbb{E}[(\hat{a}_w^{(1)}[k] - \mathbb{E}[\hat{a}_w])e_{y_2}[k]]\end{aligned}$$

the two covariances through the estimate/error cross-moment $C := \mathbb{E}[\hat{x} e^\top]$ (posterior)

$$\begin{aligned}\mathbb{E}[\delta\hat{w}_3[k-1] e_{y_1}[k]] &= \mathbb{E}[\hat{x}_3[k-1](e_1^-[k] + n_w[k])] = \mathbb{E}[\hat{x}_3[k-1] e_2[k-1]] = C_{32}[k-1] \\ \mathbb{E}[\hat{a}_w^{(1)}[k] e_{y_2}[k]] &= (C^{(1)}[k] H_2^\top)_4\end{aligned}$$

recursion of C under the filter's gains, true noise statistics ($M \neq 0$)

$$\begin{aligned}\mathbb{E}[\hat{x}[k] n_w[k]] &= (I - \ell_2 H_2) \ell_{.1}[k] R_1, \quad \mathbb{E}[\hat{x}[k] w[k]^\top] = (I - \ell_2 H_2) \ell_{.1}[k] M^\top \\ C^-[k+1] &= F C[k] F^\top + F(I - \ell_2 H_2) \ell_{.1}[k] M^\top \\ C^{(1)}[k] &= C^-[k](I - \ell_{.1} e_1^\top)^\top + \ell_{.1} e_1^\top P_{\text{true}}^-[k](I - \ell_{.1} e_1^\top)^\top - \ell_{.1} R_1 \ell_{.1}^\top \\ C[k] &= C^{(1)}[k](I - \ell_2 H_2)^\top + \ell_2 H_2 P_{\text{true}}^{(1)}[k](I - \ell_2 H_2)^\top - \ell_2 R_2 \ell_{.2}^\top \\ P_{\text{true}}^-[k+1] &= F P_{\text{true}}[k] F^\top + Q - F(I - \ell_2 H_2) \ell_{.1}[k] M^\top - M \ell_{.1}[k]^\top (I - \ell_2 H_2)^\top F^\top \\ M = 0 &\Rightarrow P_{\text{true}} = P, \quad C \equiv 0, \quad \text{both product terms} = 0 \quad (\text{innovation} \perp \text{past})\end{aligned}$$

6 Correlated-noise predict

$$\begin{aligned}w[k] &= \frac{M}{R_1} n_w[k] + \tilde{w}[k], \quad \mathbb{E}[\tilde{w} n_w] = 0, \quad \tilde{Q} := \mathbb{E}[\tilde{w} \tilde{w}^\top] = Q - \frac{M M^\top}{R_1} \\ x[k+1] &= \left(F - \frac{M}{R_1} H_1\right) x[k] + B[k] u[k] + \frac{M}{R_1} y_1[k] + \tilde{w}[k] \\ \hat{x}^-[k+1] &= F \hat{x}[k] + B[k] u[k] + \frac{M}{R_1} (y_1[k] - \hat{x}_1[k]) \\ P^-[k+1] &= \left(F - \frac{M}{R_1} H_1\right) P[k] \left(F - \frac{M}{R_1} H_1\right)^\top + \tilde{Q} \\ e^-[k+1] &= \left(F - \frac{M}{R_1} H_1\right) e[k] + \tilde{w}[k], \quad \mathbb{E}[\tilde{w} n_w] = 0 \Rightarrow C \equiv 0\end{aligned}$$

$$\begin{aligned}y_1[k] - \hat{x}_1[k] &= \frac{R_1}{S_1[k]} e_{y_1}[k] - \ell_{12}[k] e_{y_2}[k] = \mathbb{E}[n_w[k] \mid y_1[1..k], y_2[1..k]], \quad \ell_{12} = \frac{H_{24} P_{14}^{(1)}}{S_2} \\ \frac{M}{R_1} &= g_n = (1 - \lambda_c) [0, 0, -1, \bar{a}']^\top\end{aligned}$$

with the entries written out (hold; the general case adds $\Delta \bar{w}_d$ inside the bracket)

$$\begin{aligned}\delta\hat{w}_3^-[k+1] &= \lambda_c \delta\hat{w}_3[k] - (1 - \lambda_c)(y_1[k] - \delta\hat{w}_1[k]) \\ \hat{a}_w^-[k+1] &= \hat{a}_w[k] + \bar{a}'[k] (1 - \lambda_c) [\delta\hat{w}_3[k] + y_1[k] - \delta\hat{w}_1[k]] \\ &\quad + \frac{1}{2} \bar{a}''[k] (1 - \lambda_c)^2 [\delta\hat{w}_3[k] + y_1[k] - \delta\hat{w}_1[k]]^2 + (\text{mean lines}) \\ F_e(3, 1) &\rightarrow F_e(3, 1) + (1 - \lambda_c), \quad F_e(4, 1) \rightarrow F_e(4, 1) - \bar{a}' (1 - \lambda_c) \\ \tilde{Q}_{33} &= Q_{33} - (1 - \lambda_c)^2 R_1, \quad \tilde{Q}_{34} = -\bar{a}' \tilde{Q}_{33}, \quad \tilde{Q}_{44} = \bar{a}'^2 \tilde{Q}_{33}\end{aligned}$$

mean lines of 0902 under this predict ($u := \text{unknown increment}$)

$$\begin{aligned}u[k] &= (1 - \lambda_c)(e_3[k] - e_1[k]) + F_{dw}[k] e_4[k] + \tilde{w}\text{-part} \\ \mathbb{E}[e_3 u] &= (1 - \lambda_c)(P_{33} - P_{31}) + F_{dw} P_{34} \\ \mathbb{E}[u^2] &= (1 - \lambda_c)^2 (P_{33} + P_{11} - 2P_{13}) + 2(1 - \lambda_c) F_{dw} (P_{34} - P_{14}) \\ &\quad + F_{dw}^2 P_{44} + \tilde{Q}_{33}\end{aligned}$$

$$\mathbb{E}[\delta \hat{w}_3 u] = 0, \quad \mathbb{E}[\bar{a}'[k] u] = 0, \quad \mathbb{E}[\delta \hat{w}_3[k-1] e_{y_1}[k]] = 0$$

(the n_w feedthrough family of 0902 vanishes)